

MagnaTayISU

A project proposal submitted in

partial fulfillment for the subject

ITAPPDEV 3 – Mobile Applications

Prepared by

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PRODUCT REQUIREMENT DOCUMENT

1. Project overview

MagnaTayISU is a student-centered Android application for ISU students who want a simple daily record of walking activity and food intake. Built in MIT App Inventor 2, it brings step tracking, personal goals, an offline Filipino food list, and date-based reports into one app. It supports general wellness logging and is not medical or nutrition advice.

2. Users and problem addressed

The primary users are students with an Android phone and a compatible motion sensor. The app reduces reliance on separate notes by bringing activity and meal records together while keeping food calories consumed separate from walking calories burned. Routine logging is designed to work offline.

3. Functional requirements

- Dashboard: show today's steps, personal goal and progress, estimated distance, estimated walking calories burned, food calories consumed, and meals logged.
- Step Tracker: start or pause tracking, display the session and saved daily total without adding cumulative sensor readings repeatedly, and allow a confirmed reset of today without deleting earlier dates. The standard Pedometer is foreground-oriented; the separate foreground-service build still requires physical-device verification.
- Food log: search 18 starter Filipino food items, choose Breakfast, Lunch, Dinner, or Snack, enter servings greater than zero, preview calories, and add, edit, or delete entries with confirmation. Include custom foods and meal subtotals.
- History and settings: view records by date, show a seven-day walking summary, and adjust a personal step goal, stride length, and optional weight input.

4. Data, privacy, and scope

Records are stored locally in TinyDB on the device; there are no user accounts, cloud synchronization, or cross-device transfer. Distance and calorie values are estimates. The app does not use a calorie-deficit calculation, make diagnoses, prescribe weight loss, or replace professional health advice.

5. Acceptance criteria

- A fresh install opens with honest empty states and no fabricated step or food records.
- Cooked White Rice at 200 kcal per cup, entered as 2 servings, produces 400 kcal.
- Empty, invalid, zero, or negative servings are rejected; edits and confirmed deletions recalculate totals.
- Changing the selected history date does not overwrite another date; changing the goal refreshes progress.
- Step accuracy, permission behavior, and any background counting are tested on the target Android phone before they are claimed as verified.

TECHNICAL REQUIREMENT DOCUMENT

1. Platform and screen architecture

Target platform: Android. The project is authored in MIT App Inventor 2 and uses five screens: Screen1 (Dashboard), StepTracker, FoodLog, History, and Settings. Built-in components include TinyDB, Pedometer, Clock, Notifier, ListView, TextBox, and Spinner. All screens share the TinyDB namespace MagnaTayISU_DB. The background variant adds the Itoo 4.5.0 foreground-service extension; the stock project remains separate.

2. Local data model

Store settings, the seeded food list, food_entries, and date-keyed step totals in TinyDB. Each food entry has a stable unique ID and the fields [id, date, meal, food name, servings, unit, kcal per serving, total kcal]. Date keys use yyyy-MM-dd. Edit and delete operations identify the entry by its ID rather than treating a filtered ListView position as a permanent record ID.

3. Calculations and step handling

- Distance (km) = steps × stride length (cm) ÷ 100,000; the initial stride is 76 cm and can be changed.
- Walking calories are approximate: weight (kg) × distance (km) × 0.53 when a weight is entered; otherwise steps × 0.04. Food calories = kcal per serving × servings. Display consumed and burned values separately.
- The stock tracker assigns the Pedometer's cumulative session reading to sessionSteps and combines it with the saved-at-start total; it does not add the cumulative reading at every event. The background variant stores service values through Itoo properties, not TinyDB or UI blocks in the background procedure.
- The current background blocks include a heuristic filter: reject very fast events (<280 ms), ignore a single jump greater than five steps, and commit only after a 10-step bout; a gap over four seconds discards pending steps. This can miss short walks and is not validated for all phones.

4. Validation, permissions, and resilience

Validate numeric goals, stride length, servings, and custom-food calories before saving. Confirm reset and deletion. Handle empty history, missing food selection, and unavailable sensors without inventing readings. The supplied AAB manifest requests activity recognition, notifications, foreground-service/health, wake-lock, and boot-related permissions; Android prompts and battery controls vary by OS version and device.

5. Verification and limitations

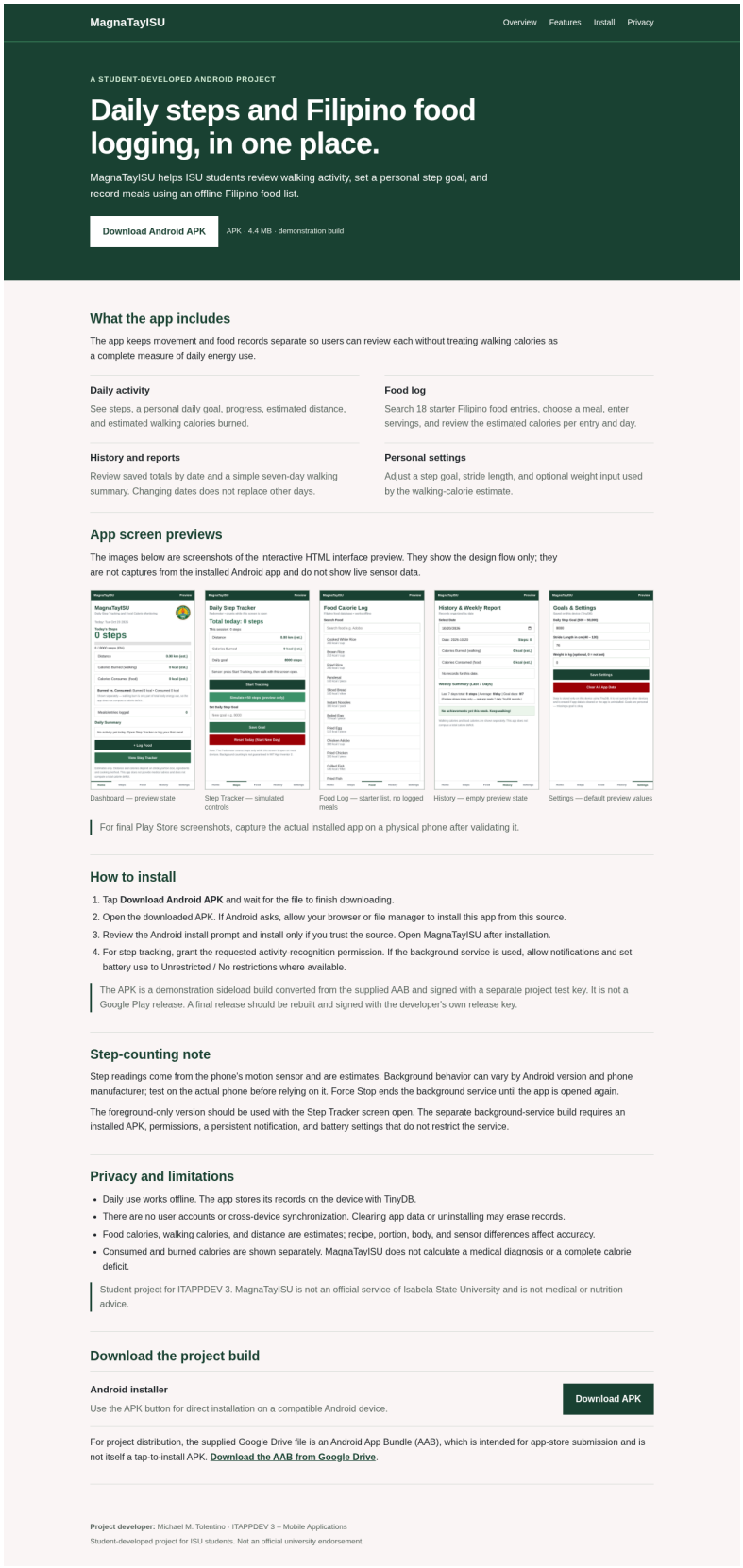
Machine-level project checks and the supplied AAB manifest have been inspected. They do not substitute for a physical-phone test. Verify sensor counts, persistence, permissions, installation, the background service, and screen-locked behavior on the final signed APK. TinyDB is device-local; values are estimates; network access is not needed for normal food logging.

User's Manual

Mobile App Assets

Webpage URL — Pending public hosting

Source files: LandingPage/index.html and MagnaTayISU.apk



MagnaTayISU

Overview Features Install Privacy

A STUDENT-DEVELOPED ANDROID PROJECT

Daily steps and Filipino food logging, in one place.

MagnaTayISU helps ISU students review walking activity, set a personal step goal, and record meals using an offline Filipino food list.

Download Android APK

APK · 4.4 MB · demonstration build

What the app includes

The app keeps movement and food records separate so users can review each without treating walking calories as a complete measure of daily energy use.

Daily activity

See steps, a personal daily goal, progress, estimated distance, and estimated walking calories burned.

Food log

Search 18 starter Filipino food entries, choose a meal, enter servings, and review the estimated calories per entry and day.

History and reports

Review saved totals by date and a simple seven-day walking summary. Changing dates does not replace other days.

Personal settings

Adjust a step goal, stride length, and optional weight input used by the walking-calorie estimate.

App screen previews

The images below are screenshots of the interactive HTML interface preview. They show the design flow only; they are not captures from the installed Android app and do not show live sensor data.

Dashboard — preview state

Step Tracker — simulated controls

Food Log — starter list, no logged meals

History — empty preview state

Settings — default preview values

> For final Play Store screenshots, capture the actual installed app on a physical phone after validating it.

How to install

1. Tap **Download Android APK** and wait for the file to finish downloading.
2. Open the downloaded APK. If Android asks, allow your browser or file manager to install this app from this source.
3. Review the Android install prompt and install only if you trust the source. Open MagnaTayISU after installation.
4. For step tracking, grant the requested activity-recognition permission. If the background service is used, allow notifications and set battery use to Unrestricted / No restrictions where available.

> The APK is a demonstration sideload build converted from the supplied AAB and signed with a separate project test key. It is not a Google Play release. A final release should be rebuilt and signed with the developer's own release key.

Step-counting note

Step readings come from the phone's motion sensor and are estimates. Background behavior can vary by Android version and phone manufacturer; test on the actual phone before relying on it. Force Stop ends the background service until the app is opened again.

The foreground-only version should be used with the Step Tracker screen open. The separate background-service build requires an installed APK, permissions, a persistent notification, and battery settings that do not restrict the service.

Privacy and limitations

- Daily use works offline. The app stores its records on the device with TinyDB.
- There are no user accounts or cross-device synchronization. Clearing app data or uninstalling may erase records.
- Food calories, walking calories, and distance are estimates; recipe, portion, body, and sensor differences affect accuracy.
- Consumed and burned calories are shown separately. MagnaTayISU does not calculate a medical diagnosis or a complete calorie deficit.

> Student project for ITAPPDEV 3. MagnaTayISU is not an official service of Isabela State University and is not medical or nutrition advice.

Download the project build

Android installer

Use the APK button for direct installation on a compatible Android device.

Download APK

For project distribution, the supplied Google Drive file is an Android App Bundle (AAB), which is intended for app-store submission and is not itself a tap-to-install APK. **Download the AAB from Google Drive**.

Project developer: Michael M. Tolentino · ITAPPDEV 3 – Mobile Applications

Student-developed project for ISU students. Not an official university endorsement.

Full-page landing-page screenshot; publish the included files to obtain the final public URL.

MOBILE APP ASSETS AND DISTRIBUTION STATUS

Google Drive file supplied by the developer: MagnaTayISU.aab

Open the supplied MagnaTayISU.aab in Google Drive

Included files

- MagnaTayISU.aia: stock App Inventor project. MagnaTayISU_Background.aia: separate Itoo foreground-service project; do not confuse the two.
- LandingPage/index.html and LandingPage/MagnaTayISU.apk: deploy together so the webpage download button resolves to the APK in the same folder.
- LandingPage/homepage_screenshot.png: full-page website capture. LandingPage/screenshots/: five separate interface captures for Dashboard, Step Tracker, Food Log, History, and Settings. These are HTML-simulation previews, not screenshots of the installed Android app; retake actual Play Store images from the final APK after device testing.

Supplied Android bundle and APK status

The shared Drive file is an Android App Bundle (AAB), version 1.0, package appinventor.ai_snipperm30.MagnaTayISU. Its manifest includes an Itoo foreground-service component. An AAB is not itself a tap-to-install APK.

The included 4.4 MB universal APK was generated from that AAB and signed with a separate project test key. It is a demonstration/sideload build, not a Google Play release and not an update signed by the original AAB key. The final APK should be rebuilt from the final App Inventor project and signed with the developer’s own protected release key.

Web and store publication status

The landing page is ready as a local static website, but no permanent public host or domain was supplied. Its public URL remains pending deployment. The app is not listed on Google Play; therefore there is no Play Store URL, final store icon set, featured graphic, or verified device screenshot set. Capture actual screens and prepare store assets only after on-device testing.

I. ADMINISTRATOR'S MANUAL

A. Prepare the project

- Use the stock MagnaTayISU.aia for foreground-only step tracking. Use MagnaTayISU_Background.aia only when the Itoo foreground service is required. Keep the two projects separate.
- Import through Projects → Import project (.aia) in MIT App Inventor. For the background project, verify the Itoo extension appears in Designer and that all Blocks work. Build an APK; the Companion is not a valid test of the background service.

B. Build and verify the release APK

- For the final public version, build the APK from the current project in the developer's App Inventor account. Test on a physical Android phone: fresh install, permissions, food calculation, persistence, date changes, goal updates, and step counting.
- For the background variant, test with the app open, another app in front, screen locked, after swipe-close, and after a Force Stop. Force Stop should stop the service until the app is opened again. Record the phone model and observed counts.
- The APK in this package is a demonstration conversion of the supplied AAB and uses a different test signing key. Android may not install it over an existing copy signed with another key. Uninstalling can erase local TinyDB records; use a fresh test device or preserve needed records first.
- Keep the production signing key private and backed up. Do not publish a test-signed APK as a production Play Store build.

C. Publish the landing page

- Upload LandingPage/index.html and MagnaTayISU.apk together to a public static host. Keep the APK filename or update both download links in index.html. Prefer HTTPS.
- Open the hosted page on an Android phone and test the APK button. Confirm the downloaded filename ends in .apk and the installer opens. Do not link the AAB while labeling it as an APK.
- After hosting, replace "Pending public hosting" in the proposal with the final public URL. Make sure the Google Drive sharing permission is set appropriately and that the linked file is actually the intended file.

D. Final submission checks

- The supplied Drive link currently points to MagnaTayISU.aab; upload the final source project, APK, AAB, proposal, and required assets to the public project folder if the instructor requires all of them there.
- The Google Play listing and actual app screenshots remain pending until the final signed app is tested and published. Do not present the HTML simulation or generated test APK as verified production evidence.

II. USER MANUAL FOR ISU STUDENTS

1. Install and open

Download the Android APK from the landing page and open it on a compatible Android phone. Android may ask you to allow your browser or file manager to install apps from that source. Review the install prompt and install only if you trust the file. The included APK is a demonstration build; use the latest developer-signed build for a final release.

2. Permissions and step tracking

Allow activity-recognition permission when requested. In the stock version, open Step Tracker and start tracking before you walk; the App Inventor Pedometer is foreground-oriented. In the background-service version, allow notifications and set the app battery use to Unrestricted / No restrictions where available. Keep the persistent notification enabled. Android brand settings can still stop a service, so verify counts on your own phone. Force Stop ends background tracking until the app is reopened.

Keep the phone in a pocket while walking for the most consistent sensor reading. The accelerometer can mistake vibration or handling for steps and can miss steps. Use Reset Today only after confirming; it resets today's total while earlier date records are intended to remain.

3. Dashboard and goals

The Dashboard summarizes today's step count, progress toward the personal goal, estimated distance, estimated walking calories burned, food calories consumed, and meals logged. To change the goal, open Step Tracker or Settings, enter a valid value, and save. Stride length and optional weight settings affect estimates; they are not medical measurements.

4. Log food

Open Food Log, search for a food, select a meal (Breakfast, Lunch, Dinner, or Snack), and enter a positive serving amount. Check the estimate, then add it to the log. Example: Cooked White Rice at 200 kcal per cup × 2 servings = 400 kcal. Use the custom-food option for an item not in the starter list and enter its own calorie estimate. Select a saved entry to edit or delete it; confirm before deletion.

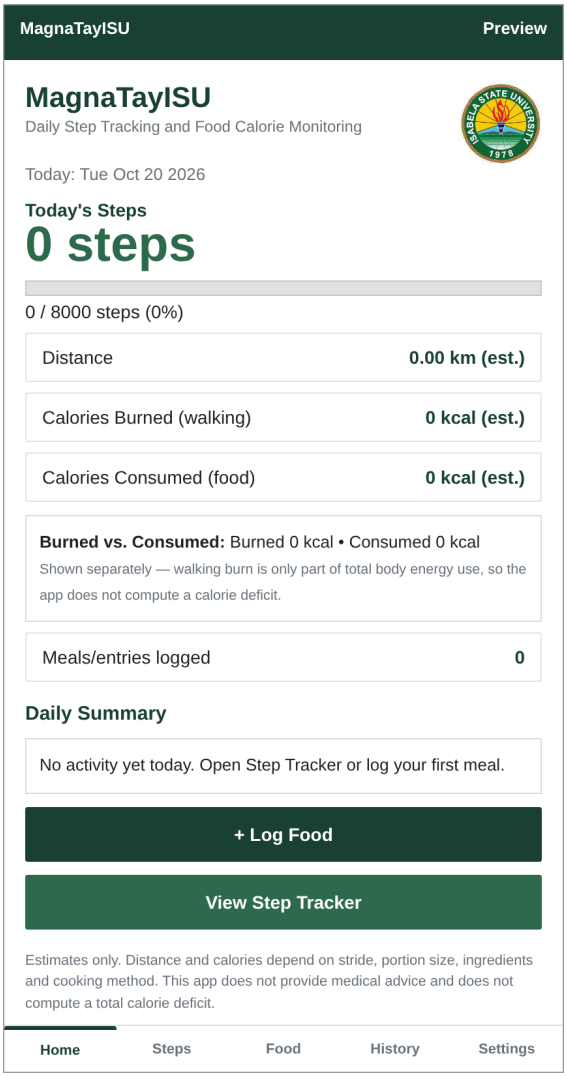
5. History, privacy, and estimates

Use History to select a date and review that day's entries and step record. The weekly report summarizes recorded walking data. Settings can adjust the goal and measurement inputs. The app stores records locally on this device; there is no account or cross-device sync. Clearing app data or uninstalling may erase records.

Food calories, distance, and walking calories are estimates. Consumed and burned calories are shown separately; the app does not calculate a complete calorie deficit or provide medical or nutrition advice.

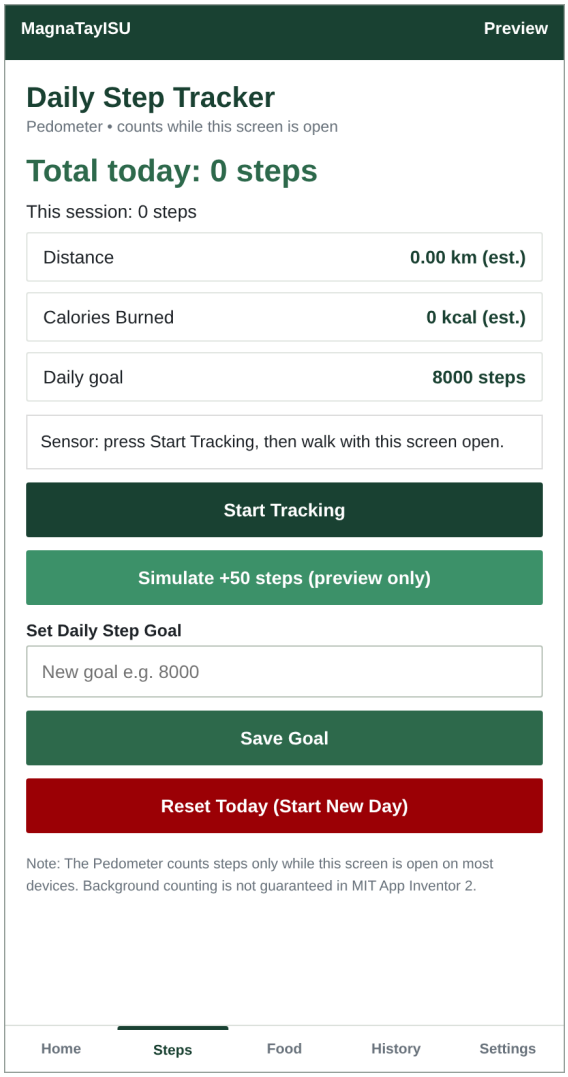
APP SCREEN PREVIEWS (1 OF 3)

HTML interface preview only; no live sensor data is shown.



Dashboard — empty preview state

Shows the dashboard layout with zero steps and no logged meals.



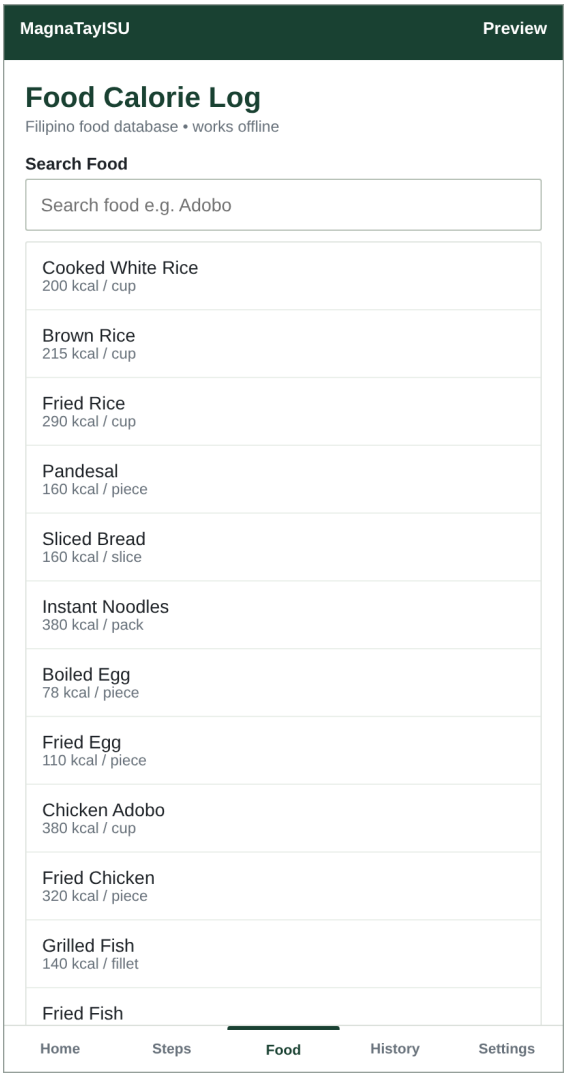
Step Tracker — simulated controls

The displayed zero count is not a physical sensor reading.

Not final Google Play assets. Capture the installed app on a phone after testing.

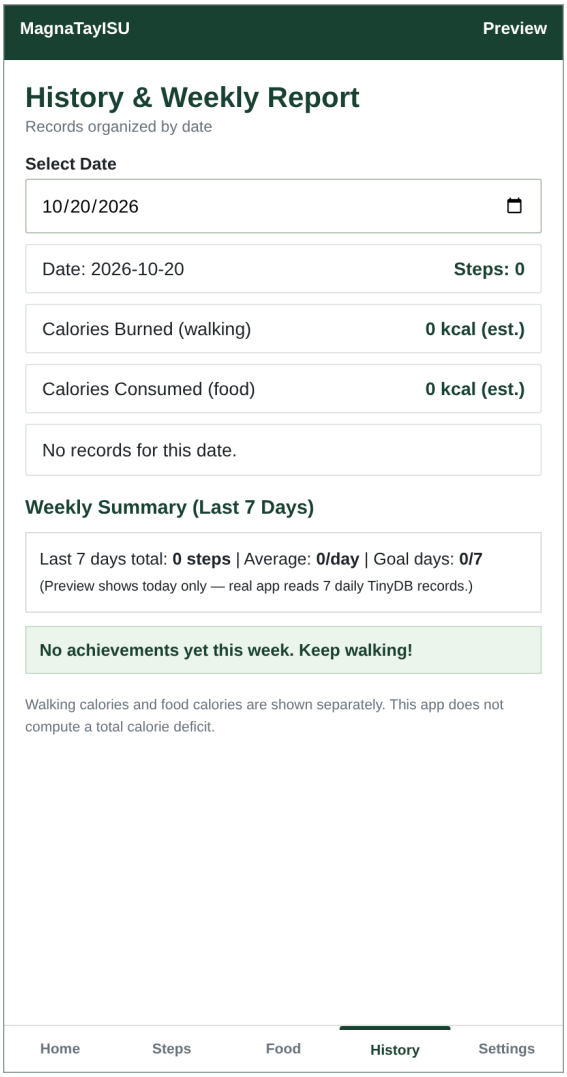
APP SCREEN PREVIEWS (2 OF 3)

HTML interface preview only; no live sensor data is shown.



Food Log — starter list

Shows food choices; no meal entry is recorded.



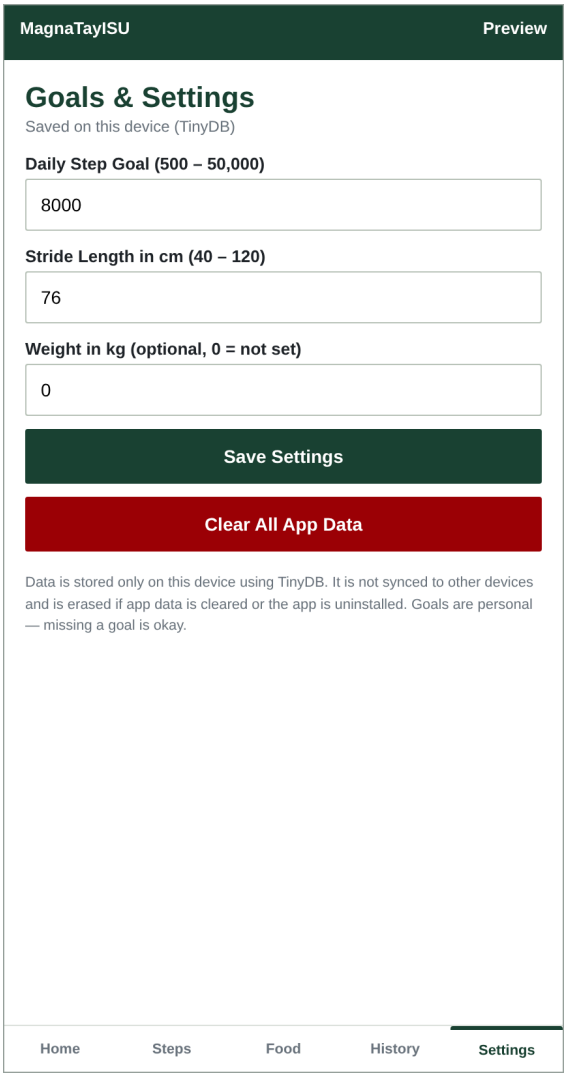
History — empty preview state

The weekly panel is a simulation, not verified app history.

Not final Google Play assets. Capture the installed app on a phone after testing.

APP SCREEN PREVIEWS (3 OF 3)

HTML interface preview only; no live sensor data is shown.



Settings — default values

Default preview inputs only; not saved phone settings.

Not final Google Play assets. Capture the installed app on a phone after testing.